

Alterations of Fractures in Carbonate Rocks by CO₂-Acidified Brines

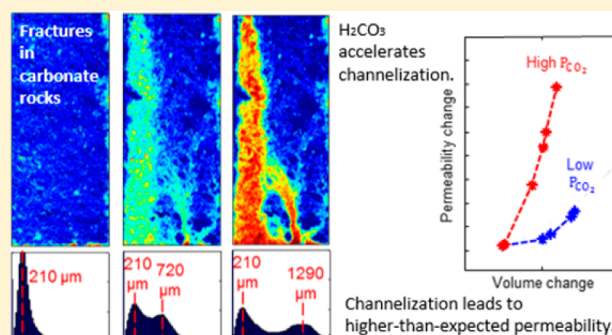
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ABSTRACT: Fractures in geological formations may enable migration of environmentally relevant fluids, as in leakage of CO₂ through caprocks in geologic carbon sequestration. We investigated geochemically induced alterations of fracture geometry in Indiana Limestone specimens. Experiments were the first of their kind, with periodic high-resolution imaging using X-ray computed tomography (xCT) scanning while maintaining high pore pressure (100 bar). We studied two CO₂-acidified brines having the same pH (3.3) and comparable thermodynamic disequilibrium but different equilibrated pressures of CO₂ (P_{CO_2} values of 12 and 77 bar). High- P_{CO_2} brine has a faster calcite dissolution kinetic rate because of the accelerating effect of carbonic acid. Contrary to expectations, dissolution extents were comparable in the two experiments. However, progressive xCT images revealed extensive channelization for high P_{CO_2} , explained by strong positive feedback between ongoing flow and reaction. The pronounced channel increasingly directed flow to a small region of the fracture, which explains why the overall dissolution was lower than expected. Despite this, flow simulations revealed large increases in permeability in the high- P_{CO_2} experiment. This study shows that the permeability evolution of dissolving fractures will be larger for faster-reacting fluids. The overall mechanism is not because more rock dissolves, as would be commonly assumed, but because of accelerated fracture channelization.



INTRODUCTION

Carbonate minerals are abundant in geological settings and are highly relevant to activities such as oil and gas extraction,^{1–3} waste injection,⁴ aquifer recharge,⁵ and geologic carbon sequestration.^{6,7} Their dissolution is promoted when acids are introduced or formed from injected fluids. Of the predominant carbonates, calcite is the most soluble and fast-reacting. Because calcite is abundant, its dissolution can substantially alter subsurface geologic media and the migration of environmentally relevant fluids.⁸ Furthermore, calcite dissolution can lead to the release of heavy metals and cause freshwater and soil contamination.^{9–11} Hence, calcite reactions are important determinants of the risks to subsurface systems posed by exposure to acidic fluids.

According to transition state theory (TST),^{12–16} the surface-area normalized dissolution rate of calcite, R [mol m⁻² s⁻¹], is

$$R = k \left(1 - \frac{a_{\text{Ca}^{2+}} a_{\text{CO}_3^{2-}}}{K_{\text{sp}}} \right) \quad (1)$$

The term in brackets is the thermodynamic driving force, which is a measure of how far away the system is from equilibrium, in which a_i is the activity of species i , and K_{sp} is the solubility product of calcite. The kinetic coefficient, k , is the net of the forward and reverse rates of three parallel dissolution mechanisms:^{12,17}

$$k = k_1 a_{\text{H}^+} + k_2 a_{\text{H}_2\text{CO}_3^*} + k_3 a_{\text{H}_2\text{O}} - k_{-3} a_{\text{Ca}^{2+}} a_{\text{CO}_3^{2-}} \quad (2)$$

At low pH, the hydrogen ion term is often dominant.^{12,17} However, for conditions of geologic carbon sequestration, the carbonic acid term may be comparable in magnitude because equilibration with high-pressure CO₂ results in pH as low as 3 and H₂CO₃^{*} concentrations as high as several mol/L.

Previous studies have demonstrated that fast dissolution of calcite can lead to substantial change in rock porosity and permeability in a short timeframe.^{6,15,16,18–25} In fractures, such alterations are of special interest because fractures dominate flow and solute transport.^{26–33} Furthermore, under flow regimes with relatively high Damkohler (Da) and Peclet (Pe) numbers, wormholes or channels develop in fractures^{22,34–38} and can cause further permeability enhancement.³⁹ Lacking are experimental investigations on how variation in carbonate solution chemistry may affect these processes.

The primary objective of this study was to investigate geochemically driven alterations in fractures in carbonate rocks, with a focus on how variation in carbonate solution chemistry and reaction kinetics shape fracture geometry and permeability. Toward this goal, high-pressure flow-through experiments were

Received: April 19, 2015

Revised: July 22, 2015

Accepted: July 23, 2015

Published: July 23, 2015

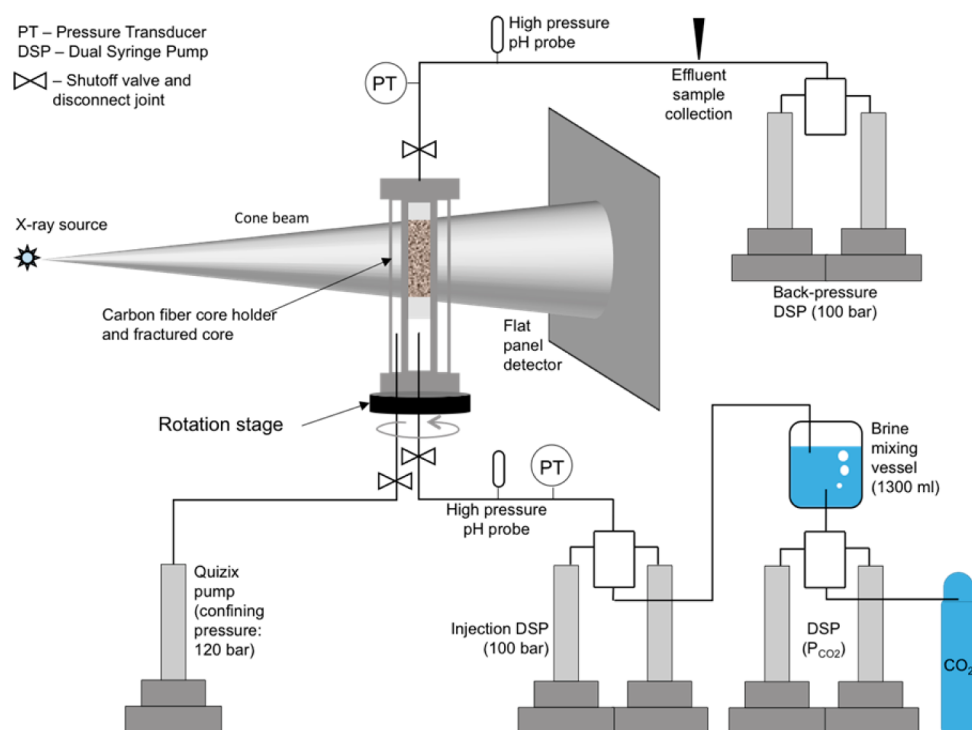


Figure 1. Experimental setup showing the high-pressure flow-through system with the core holder on the rotation stage of the xCT scanner.

conducted under conditions relevant to the deep subsurface environment. A total of two brines were investigated. They were equilibrated with CO_2 at different pressures, 12 and 77 bar P_{CO_2} , and the compositions were slightly different to achieve the same pH (3.3). The flow experiments were conducted at 100 bar to represent conditions relevant to geologic carbon sequestration. The high- P_{CO_2} brine is representative of CO_2 -acidified brine near a deep CO_2 -injection well, and the low- P_{CO_2} brine represents a more diluted CO_2 -acidified brine. The resulting calcite saturation indices ($\text{SI} = \log(a_{\text{Ca}^{2+}}a_{\text{CO}_3^{2-}}/K_{\text{sp}})$) of the two brines are different, but both solutions are so far from equilibrium that in each case that the thermodynamic driving-force term (shown in eq 1) is approximately unity. Therefore, we were able to isolate the kinetic effects of the carbonic acid concentration (shown in eq 2).

This investigation is also unique in that the experimental setup advances the ability to characterize the temporal evolution of fracture geometry and permeability. It is the first of its kind in coupling a high-pressure flow-through system with high-resolution X-ray computed tomography (xCT) imaging during the experiment. One of the challenges we addressed is the rotational scanning of the specimen while maintaining both the high pore pressure and the confining pressure of the core holder. Similar systems in prior experiments used medical scanners for periodic real-time imaging.^{6,40–44} The resulting images, with a resolution on the order of 100 μm , provided valuable insights on the rough progression of matrix porosity. However, for accurate geometry quantification of fine-scale features such as fractures under high confining pressure, higher resolution is needed.⁴⁵ The unique and detailed data set of the xCT images generated in this study, with a resolution of approximately 30 μm , permits the making of unique insights on fracture alterations caused by calcite dissolution.

The formation selected for this study is Indiana Limestone. It is a homogeneous carbonate widely used as a model material for research purposes, and its hydrologic properties have been characterized.^{46,47} This rock was selected because of its uniformity and consistency, which allowed us to isolate the effects of brine chemistry. Furthermore, these experiments enable a direct comparison with previous experiments that studied mineralogically heterogeneous limestone rock.^{6,43,48}

In addition to the effluent chemistry and the xCT results for two fracture-flow experiments, the results from two types of numerical simulation are reported. For each fracture geometry, variable aperture maps were first constructed and used in a 2D steady-state flow simulation to estimate the fracture permeability, which is a critical parameter in determining fluid flow in fractured materials. Second, a 1D nonsteady-state reactive-transport model was used as a theoretical benchmark to examine the role of kinetic limitations in calcite dissolution. Because there were no published kinetic rate parameters applicable to high pressures representative of the deep subsurface, a new set of kinetic rate parameters was determined.

■ MATERIALS AND METHODS

Fracture Flow Experiments. The core fracturing and preparation procedures are detailed in the [Supporting Information](#). The cores were 1 in. in diameter and 2 in. in length. [Figure 1](#) shows a schematic of the coupled core-flow and xCT imaging experimental setup at the National Energy Technology Laboratory (NETL) in Morgantown, West Virginia. The following details can be found in the [Supporting Information](#): flow rate and pressure control, temperature control options, pH measurement, effluent sample collection, and effluent chemistry analysis.

The core-holder was fixed to the rotation stage of the xCT scanner (North Star Imaging M-5000). The core was scanned under pressure before reactive flow and at time points during

Table 1. Influent Compositions of the Two Brines and the Estimated Kinetic Coefficients and Equilibrium Concentrations

	brine recipe		influent brine chemistry					kinetic coefficient ^a	after equilibration with calcite ^b	
	NaCl [M]	CaCO ₃ [M]	total carbon [M]	H ₂ CO ₃ * [M]	pH	ionic strength [M]	SI calcite	k (mol/m ² s)	Ca ²⁺ [M] ^c	pH
low- <i>P</i> _{CO₂} 12 bar	1	0	0.303	0.303	3.3	0.96	−∞	0.9 × 10 ^{−4}	0.0266 (1066)	5.4
high- <i>P</i> _{CO₂} 77 bar	1	0.001	1.12	1.12	3.3	0.94	−5.15	2.0 × 10 ^{−4}	0.0419 (1678)	5

^aKinetic coefficient based on the influent chemistry using the kinetic parameters fitted for this study. ^bExpected conditions for calcite equilibration with $K_{sp} = 3.311 \times 10^{-9}$. ^cValues in parentheses have units of mg/L to be consistent with Figure 3.

the experiments, as detailed in the [Supporting Information](#). To accomplish rotation during a scan, the flow was stopped and the core-holder disconnected and sealed to ensure that the core did not experience a change in pore or confining pressure. During the methods development phase, attempts were made to conduct xCT scans while maintaining the flow connection by using a rotating union. However, when the core-holder was rotated, the torque at the connection created vibrations that undermined the image quality.

The resulting 3D volume images of the cores had optimized voxel sizes of 28 and 30 μm for the low- and high-*P*_{CO₂} experiments, respectively. Image processing of the reconstructed 3D xCT image, to isolate the fracture and distinguish it from the rock matrix, was especially challenging because of very small apertures and large aperture variations. To address this, we developed a novel segmentation routine.⁴⁵ The method, called Technique of Iterative Local Thresholding (TILT),⁴⁹ is fully automated and computationally efficient. The output of the segmentation routine is a 3D volume of the fracture. For this work, the volume image was converted into a 2D map of the apertures of the fracture. The resulting fractures had initial average apertures of approximately 200 μm under pressure. (For additional aperture statistics, see [SI-Table 2](#).)

To synthesize two brines with different *P*_{CO₂} conditions but the same pH, we used PHREEQC⁵⁰ to determine the CaCO₃ needed in the influent brine (shown in [Table 1](#)). Also shown is the estimated kinetic coefficient *k* as well as the Ca concentrations and pH expected upon equilibrium with calcite. The value of *k* for the high-*P*_{CO₂} condition is twice as large as that for the low-*P*_{CO₂} condition because of the greater contribution of the H₂CO₃* term (eq 2). The high-*P*_{CO₂} brine has a higher predicted equilibrium Ca concentration and slightly lower pH because it requires greater calcite dissolution to reach equilibrium.

Flow Simulations. Here, two-dimensional steady-state flow simulations using boundary conditions of a constant pressure gradient were conducted to quantify fracture permeability and its evolution. For each fracture at each point in time, the 2D aperture map was used to construct a grid in which each cell has a unique aperture. No downscaling was performed on the aperture maps, i.e., the grid resolution is the same as the 3D xCT image resolution. The 2D local cubic law (LCL) model was adopted.^{48,51–54} In each cell, the parallel-plate cubic law is assumed valid, and the depth-averaged Reynolds equation was solved using the five-point central finite difference numerical scheme.⁵⁵ At each cell edge, the harmonic mean of the apertures of the adjacent cells is used for local permeability calculation.⁵¹ The velocity and pressure were resolved in each grid cell. The resulting pressure field was then used to calculate permeability for the fracture. To visually depict flow in the

fracture, the velocity field was used to simulate tracer transport. To match the experimental conditions, a constant volumetric flux at the inlet boundary and constant pressure at the outlet were imposed.

Reactive-Transport Modeling. As a theoretical benchmark, a 1D nonsteady-state reactive-transport model was constructed in which the fracture was discretized along the flow direction, *x*. The transverse dimension is assumed to be spatially uniform. Each grid cell (Δx) is approximately 300 μm , 10 times the image resolution. The initial aperture in each cell is the average of the corresponding transverse section of the 2D aperture map of the unreacted fracture. Therefore, the 1D reactive-transport model captures the aperture variations along the flow direction as well as the variations in solution chemistry, which can be considerable given the length of the fracture.⁵⁶

In each cell, the components Ca and total carbon were modeled using the advection–diffusion–reaction equation:

$$\frac{\partial(bC)}{\partial t} = -\nabla \cdot (\vec{q}C) + \nabla \cdot (bD \cdot \nabla C) + AR \quad (\text{eq 3})$$

where *b* is the aperture, $\vec{q}$ is volumetric flow rate, and *C* denotes the component concentration. The concentrations of other species (H₂CO₃*, HCO₃[−], CO₃^{2−}, and H⁺) were calculated based on the dissociation of carbonic acid and the charge balance for the reactions detailed in [SI-Table 3](#). The diffusion coefficient *D* was set at $1 \times 10^{-9} \text{ m}^2/\text{s}$.⁵⁷ The reaction term was solved using eq 1. The surface area, *A*, for each grid cell is the nominal surface area of the cell, i.e., twice the width of the fracture multiplied by Δx . In reality, because of roughness, the actual surface areas ([SI-Table 2](#)) are approximately 30% higher than the nominal surface areas, as evidenced by the xCT data. However, using the nominal surface area is justified for fractures with surface area 1–2 times the nominal surface area.⁵⁸ The surface area was set as constant throughout the simulation.

Flow, boundary, and initial conditions are detailed in the [Supporting Information](#), along with numerical solution methods. The effective permeability for the fracture was calculated using Darcy's equation and individual permeabilities at each slice, for which the cubic law was applied.⁵⁹

For this work, a new set of kinetic parameters was determined by fitting published rate data for *P*_{CO₂} up to 50 bar.^{13,14} The statistical regression approach and the resulting parameter estimates are detailed in the [Supporting Information](#). Additionally, two published parameter sets were examined for the purposes of comparison and analysis of uncertainty. The Chou et al. (1989)¹² set of parameters, which are based on data at low pressures, represents the most widely applied parameters for calcite dissolution kinetics. The set determined for this study and that published by Molins et al. (2014)⁶⁰ were both determined by fitting to high-pressure experimental data from Pokrovsky et al. (2005 and 2009)^{13,14} but for different pressure ranges. There are similarities in these two sets ([SI-Table 4](#)), but

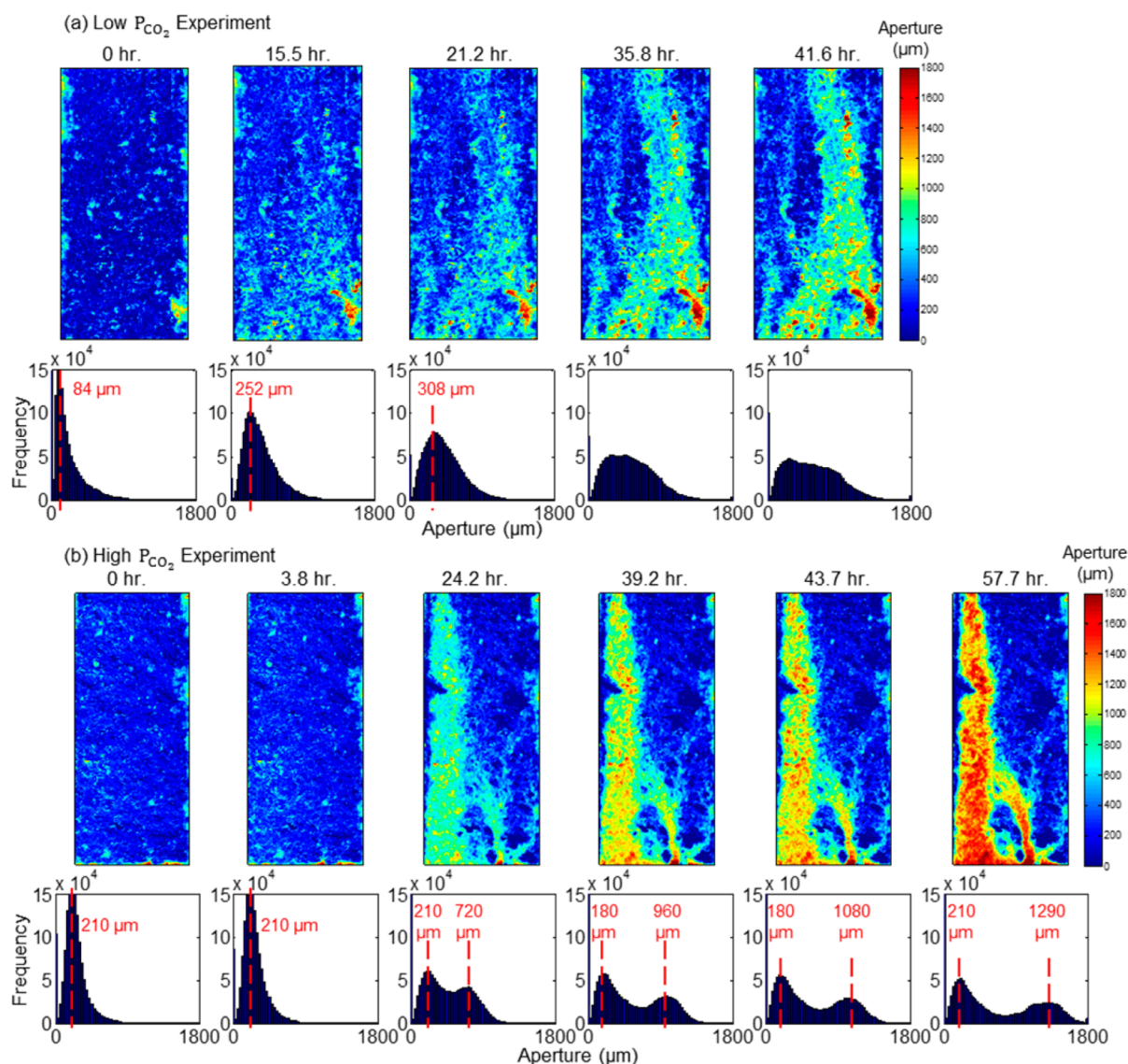


Figure 2. Aperture maps and aperture histograms at different time points for the (a) low- and (b) high- P_{CO_2} experiments. The dimensions of these images are (a) $2.38 \times 5 \text{ cm}^2$ and (b) $2.34 \times 5.5 \text{ cm}^2$. The bottom of the image is the location of the inlet. The apertures included in the histograms do not include the zero apertures, characterized as contact area (see SI-Table 1). The values indicated on the histograms are the modes of the distributions.

the difference in the k_2 parameters is statistically significant. Comparing these two sets with the Chou parameters, the hydrogen ion parameter is an order of magnitude smaller, and the carbonic acid parameter is smaller by a factor of 3.

RESULTS AND DISCUSSION

Fracture Geometry Evolution. The reaction extent and rate were examined using the xCT data by calculating the fracture volume as the sum of the voxels characterized as fracture. Because k is substantially higher for the high- P_{CO_2} experiment, it was expected that the extent of the dissolution should be larger. This is given that all other factors, such as flow rate and initial surface areas, were comparable in the two experiments. However, the observed changes in fracture volumes show otherwise. Over the entire experiment (42 h), the low- P_{CO_2} experiment fracture increased by 0.42 cm^3 in volume (186%). For the high- P_{CO_2} experiment (58 h), the increase was 0.44 cm^3 (141%). In other words, the low-and

high- P_{CO_2} experiments had nearly identical calcite depletion rates of 1.0×10^{-2} and $0.8 \times 10^{-2} \text{ cm}^3/\text{h}$, respectively.

The dissolution extents were comparable in the two experiments regardless of the variations in the influent chemistry because the dissolution patterns were different. Figure 2 shows the aperture maps and aperture distributions at different time-points during the two experiments. For the low- P_{CO_2} experiment, the dissolution pattern in the fracture is relatively uniform, although an emerging channel can be discerned after 35.8 h. The aperture distribution shifts to the right and is widened over time, but the unimodal shape is preserved. In contrast, the aperture maps of the high- P_{CO_2} experiment prominently reveal the phenomenon of channelization. In accordance with the development and deepening of the channel, the aperture distribution transforms from unimodal to bimodal as early as 24.2 h. The peak position of the fine apertures remains almost constant, while the peak position of

the large apertures continually shifts to higher values as the channel deepens.

The presence of channels as observed in the experiments is consistent with previous studies of reactive flows in fractures. First, the estimated Da (approximately 0.1) and Pe (approximately 300) are within the reported ranges for channelization.^{22,37} Second, the initial fracture apertures are heterogeneous and can trigger local flow and reaction perturbations for channel initiation. However, the relative strength of channelization deviates from what is expected on the basis of the statistical characteristics (SI-Table 2). The initial fracture of the low- P_{CO_2} experiment has twice-as-large roughness and contact area, a smaller average aperture, and a larger correlation length in both the longitudinal and the transverse directions. Even with these favorable features,^{37,38,61,62} this fracture did not develop a stronger channel.

The discrepancy in channelization strength between the two experiments can be explained by differences in the influent chemistry. Development of a channel requires the amplification of the initial perturbations, which relies on the positive feedback between flow and reaction, created by mineral dissolution.^{61,62} Interferences with this feedback suppress wormholing.⁶³ In these experiments, the dissolution rates were dependent on the surface-reaction kinetics. (The average flow velocities, approximately 0.2 cm/s as determined by the flow simulations reported below, were orders of magnitude higher than reported values for transport-controlled reactions.⁶⁴) For the high- P_{CO_2} experiment, because of the larger kinetic coefficient and faster localized dissolution, the positive feedback between flow and reaction was accelerated. As the channel deepened, the reactive flow was increasingly focused there. Similar findings have been documented for wormholing in porous media in different P_{CO_2} conditions.¹⁵

Although our experiments were conducted under low temperature relative to the deep subsurface, this finding would extend to high-temperature conditions. In fact, the effect of the influent chemistry is expected to be stronger under higher temperatures. Because the kinetic coefficients increase significantly with temperature,⁶⁵ given the same contrast in brine compositions, the difference in the kinetic rates would be even larger.

Effluent Chemistry and Reactive-Transport Modeling.

The extent and rate of calcite dissolution were also examined using the effluent Ca concentrations. Figure 3 shows that the low- P_{CO_2} experiment produced effluent Ca concentrations that ranged from 350 to 720 mg/L and averaged 480 mg/L, and the high- P_{CO_2} experiment produced similar effluent concentrations that ranged from 350 to 500 mg/L and averaged 394 mg/L. Neither experiment reached calcite equilibrium. Continuous disequilibrium is also consistent with effluent pH lower than equilibrium and negative calculated calcite saturation indices as shown in SI-Figure 2. This confirms that the calcite dissolution reaction in the fracture was kinetically limited. To explore this, we use the 1D reactive-transport model as a theoretical benchmark.

To start, we assess the uncertainties that derive from the selection of kinetic coefficients. The 1D model results using the three sets of kinetic parameters are plotted in Figure 3. In all cases, the predicted effluent Ca concentrations rapidly reach steady-state values. The Chou parameters, which are not intended for high-pressure conditions, result in negligible

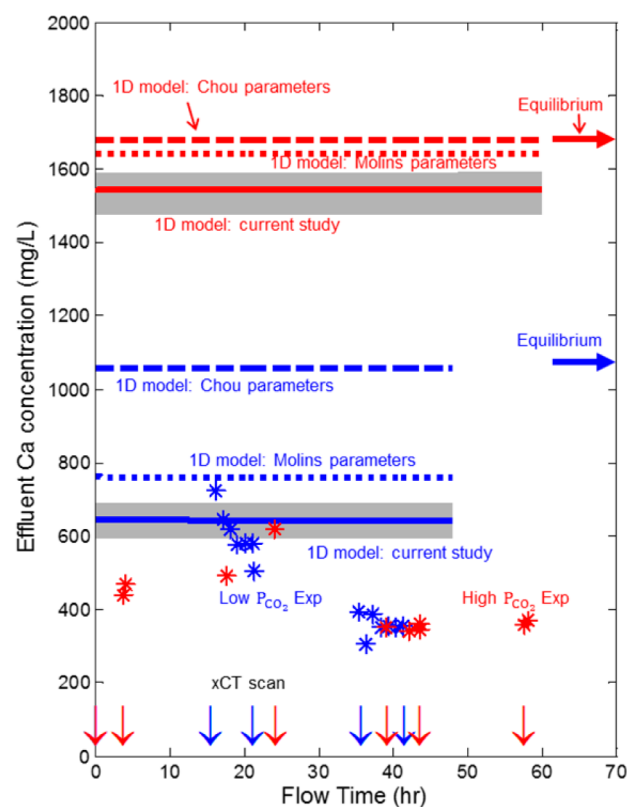


Figure 3. Effluent Ca concentrations from experimental measurements (* symbols) and 1D reactive-transport simulations using different kinetic parameters (lines). The simulation results using the parameters determined for this study are shown in the shaded bands, illustrating the range of predicted Ca concentrations using kinetic parameters within 95% confidence intervals. All of the blue symbols and lines are for the low- P_{CO_2} experiment, and all of the red symbols and lines are for the high- P_{CO_2} experiment. The downward arrows indicate when xCT scans were performed. The time axis represents the experimental flow time and does not include the 1.5 h time period for each scan.

kinetic limitations and predict effluent concentrations nearly identical to calcite saturation values. The other two parameter sets predict kinetically limited calcite dissolution, with the parameters determined in this study predicting the greatest kinetic limitations. It is interesting that at the higher P_{CO_2} , the kinetic limitations are diminished, but at intermediate CO_2 pressures (approximately 10 bar), kinetic limitations can be substantial.

Now, we compare the experimental measurements with the 1D model predictions to examine the extent to which the presence of channels affects the transverse uniformity assumption. For this, we use the kinetic parameters determined for this study. For the low- P_{CO_2} conditions, the 95% confidence interval predictions of effluent Ca concentrations overlay some of the experimental observations. The measurements are reasonably in agreement with the 1D model estimates, given the model uncertainties and experimental error. In comparison, for the high- P_{CO_2} conditions, the 95% confidence interval predictions represent a significant overestimation relative to the effluent observations. This holds regardless of the kinetic parameters used. This also means an overestimation in the bulk dissolution rate, which is proportional to the bulk reaction rate. The 1D model estimates that the higher concentration of $H_2CO_3^*$ would result in a bulk reaction rate approximately

twice as high as that of the low- P_{CO_2} experiment. However, the bulk reaction rates inferred from the measurements (based on the average effluent Ca concentrations) are comparable: 1.1×10^{-2} and 0.9×10^{-2} cm^3/h for the low- and high- P_{CO_2} experiment, respectively (which are in agreement with the estimates from the xCT data presented earlier).

The comparisons between the experimental results and 1D simulation results illustrate that kinetic limitations alone are insufficient to explain the observed extent and rate of calcite dissolution. The nonuniform fracture geometry, which resulted from channelization, altered hydrodynamics to such an extent as to invalidate the 1D reactive-transport model's assumption of transverse uniformity. For fractures with a less-developed channel and relatively low aperture variance, such as the fracture of the low- P_{CO_2} experiment, the assumption of complete transverse mixing holds.

Flow Evolution and Permeability. To visualize how channels affect fluid flow and solute transport, tracer simulations were conducted on the after-reaction fracture geometry (57.7 h) of the high- P_{CO_2} experiment. A conservative tracer of 1 mol/L was continuously released at the inlet into a steady-state flow field using the 2D model. Figure 4 shows a

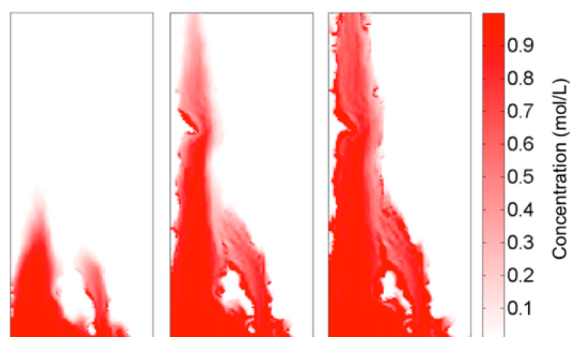


Figure 4. Snapshots of tracer transport simulations in steady state flows for the final fracture geometry of the high- P_{CO_2} experiment. The inlet is at the bottom. [Animation of tracer simulation](#) is available online.

snapshot sequence of the simulated tracer concentration. ([Animation of tracer simulation](#) is available online.) As expected, the distribution of the tracer follows the channels seen in the aperture map. Examination of the numerical output indicates that the flow velocities in the channelized part of the fracture (in general, $\geq (1 \times 10^{-3})$ m/s) are higher than the flow velocities in the nonchannelized regions (in general, $\leq (1 \times 10^{-5})$ m/s). This helps our understanding of what happens in the context of reactive flow, in which the fracture surface within the channel is in contact with a continued supply of fresh reactive fluid and thus promotes further reaction. In contrast, the fracture surface outside the channel, although in contact with fluid and available for reaction, is increasingly inaccessible to fresh reactive flow. As a result, the net dissolution across the whole fracture is reduced. This explains the experimental findings (both xCT and effluent chemistry) that the observed reaction rate of the high- P_{CO_2} experiment is much lower than what is expected. This also explains why, as the preferential flow paths deepened over time (Figure 2), the overall dissolution rate decreased slightly, as observed in the effluent Ca concentration measurements (Figure 3).

Now we examine how channelization affects fracture permeability. Figure 5 shows the permeability evolution in

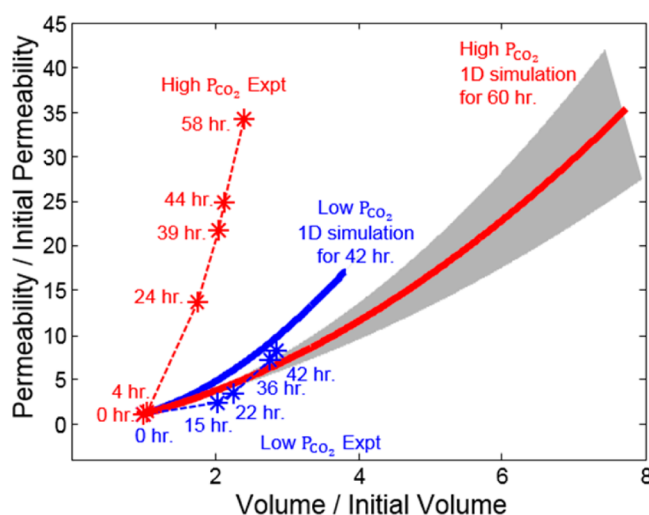


Figure 5. Permeability increase in relation to the fracture volume increase for the low- P_{CO_2} experiment (blue) and the high- P_{CO_2} experiment (red) from xCT images (dashed line with stars) and 1D simulations (solid line with gray shaded band representing the 95% confidence range of estimated kinetic parameters).

relation to the increases in fracture volume. The increases in permeability are substantial, with an increase of more than an order of magnitude in the high- P_{CO_2} experiment. Of course, this experimental system is small, and over a longer length scale, pH buffering would slow the rate of reaction and the development of channels. However, the shape of the curve suggests an ever-increasing rate of permeability increase for a given unit length.

The extents of permeability increase differ dramatically between the two experiments. For the high- P_{CO_2} experiment, the fracture permeability increased a factor of 35 at the end of the experiment, and the fracture volume increased by a factor of 2.5. In contrast, at the end of the low- P_{CO_2} experiment, the permeability increase was less than a factor of 10, and the fracture volume increased by a factor of approximately 3. The divergence of the two curves suggests an ever-increasing difference between the two scenarios.

Figure 5 also shows permeabilities predicted by the 1D reactive-transport model. For the low- P_{CO_2} experiment, the permeability–volume change relationship predicted by the 1D model is in agreement with that indicated by the geometry measurements. However, for the high- P_{CO_2} experiment, the 1D model fails to capture the relationship because it largely overestimates the dissolution and fracture volume increase. The divergence between the observed trend and the 1D prediction implies that as the channel deepens over time, for the same volume amount of dissolution, the 1D model would underestimate the fracture permeability by an ever-increasing amount.

ENVIRONMENTAL IMPLICATIONS

These findings contribute to an emerging body of work demonstrating how geochemical processes can rapidly alter fracture flow in carbonate rocks and potentially compromise subsurface storage and energy-extraction operations. Acidic erosion of calcite can substantially increase fracture volume, and

any resulting increase in permeability could lead to unexpected and unwanted migration of CO₂, brine, and associated contaminants. Dissolution that creates channels in fractures is especially important because channels can accelerate this permeability change. Furthermore, channels can provide lasting flow conduits even when large-scale compressive stress would otherwise lead to fracture closure.³⁸

Fracture permeability evolution in the field may be inferred from estimates of fracture volume change based on changes in fluid chemistry. For parallel-wall fractures, the hydraulic aperture, which is proportional to the square root of the fracture permeability, increases linearly with fracture volume. This study has shown that, for channelized fractures, using the fracture volume change as an indicator for fracture permeability change would result in an inaccurate prediction by orders of magnitude. Supposing such estimates are used as inputs for large-scale simulations, the errors will propagate, resulting in underestimation of migration of environmentally relevant fluid such as fracking fluid and CO₂ brine.

There is significant complexity in how a channel evolves in relation to reaction kinetics, and this work significantly advances our understanding of the key role played by high-pressure carbonic acid. Once a channel is initiated, the focused flow of reactive fluid will cause deepening of the channel, which will reinforce the channelized flow. This positive feedback will be stronger in the presence of fluids with faster kinetics, leading to pronounced channelization. The presence of strong channels limits the accessibility of the fracture surface to the reactive fluid and thus reduces overall reaction. Failing to account for channelization would result in large overestimation of the extent of dissolution.

In high-pressure systems involving CO₂, it is necessary to consider the catalytic effect of not only hydrogen ions but also carbonic acid. Under conditions that are relevant to geologic carbon sequestration, the accurate quantification of the catalytic effects requires that kinetic parameters based on experimental data of similar conditions be used. One of the contributions of this work is a new set of calcite reaction kinetic parameters applicable at high carbonic acid concentrations.

The findings of our study also provide insights into how fracture channels would propagate over long length scales in carbonate rocks. Because a reactive fluid with faster kinetics can result in a reduced overall reaction, the reactivity of the fluid along the fracture is less affected by pH buffering and would persist over a longer length scale. It stands to reason that the fluid with faster kinetics will produce longer channels and increase the likelihood of breakthrough into a high permeability formation. For example, in the context of geological carbon sequestration, the likelihood of breaching a confining unit would be higher.

At the reservoir scale, there is a need for practical lower-order models, such as those that form the modules in the U.S. Department of Energy's National Risk Assessment Partnership (NRAP) model for risk assessment in geologic carbon sequestration.⁶⁶ This work has shown that simple 1D models may be adequate if channels do not form. However, in a strongly channelized fracture, a 1D reactive-transport model is insufficient to predict dissolution and fracture permeability evolution. Either 2D simulations or modifications of the 1D model, such as using an effective surface area, are needed to account for channelization.

Permeability evolution trajectories from previous fracture flow studies on different rock samples are in contrast with the

steep increase observed in this study for a channelized fracture. For instance, in mineralogically heterogeneous limestone rocks, preferential dissolution of a fast-reacting mineral such as calcite can lead to less-than-expected permeability increase.^{6,48} Similarly, if the fast-reacting mineral is not distributed contiguously along the flow path, channelization will be restricted, preventing a substantial fracture permeability increase that may otherwise occur.⁶⁷ In contrast, if particles of less-soluble minerals are released as a result of preferential dissolution, they may clog flow paths and lead to permeability decrease.^{43,68} Permeability reduction can also occur in fractures primarily composed of less reactive minerals as geo-mechanical forces dominate.⁶⁹

These studies collectively demonstrate that geochemical processes are critical controlling factors for fracture permeability evolution and, therefore, need to be considered in permeability projection, especially for reservoir scale models that rely on simple porosity–permeability relationships.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.5b01980.

Indiana Limestone core fracturing and preparation procedures. Details of the high-pressure core-flow experiments and xCT scanning procedures. Statistical characterization of the initial fracture geometries. Details of the 1D reactive-transport modeling. Details of the estimation of the kinetic parameters. Effluent chemistry data including pH and saturation indices. (PDF)

● Web-Enhanced Feature

Animation of tracer simulation is available in the HTML version of the paper.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was funded by the National Science Foundation (NSF) grant CBET-1133849. H.D. acknowledges additional support by an appointment to the U.S. Department of Energy (DOE) Postgraduate Research Program at NETL administered by ORISE. We also acknowledge the use of the ICP-OES facility in Dr. Higgins' lab in the Department of Geosciences at Princeton University. Finally, we acknowledge the reviewers for their detailed and thorough assessments, which were extremely helpful in improving this manuscript.

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